

AI in Software Development – Assessment Questions

1. Short Answer Questions

Q1: Explain how AI-driven code generation tools (e.g., GitHub Copilot) reduce development time. What are their limitations?

Answer:

AI-driven code generation tools like GitHub Copilot reduce development time by:

- Automatically suggesting code completions based on context, reducing the amount of code developers need to write manually
- Generating boilerplate code and repetitive patterns instantly
- Providing quick solutions to common programming tasks
- Helping developers learn new APIs and frameworks through intelligent suggestions

Limitations:

- May generate incorrect or insecure code that requires human review
 - Can suggest outdated or inefficient solutions
 - Lacks understanding of project-specific business logic and requirements
 - May introduce licensing concerns if trained on copyrighted code
 - Cannot replace human judgment for architectural decisions
 - May reduce code quality if developers rely on it without understanding the generated code
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Q2: Compare supervised and unsupervised learning in the context of automated bug detection.

Answer:

Supervised Learning for Bug Detection:

- Requires labeled training data with examples of bugs and non-bugs
- Learns patterns from known bug examples to identify similar issues
- More accurate when sufficient labeled data is available

- Examples: Detecting specific bug types like null pointer exceptions or memory leaks based on historical bug reports
- Limitation: Requires extensive manual labeling effort and may miss new bug types not in training data

Unsupervised Learning for Bug Detection:

- Does not require labeled data; identifies anomalies and unusual patterns
- Detects deviations from normal code behavior or structure
- Can discover unknown bug types and edge cases
- Examples: Identifying unusual code patterns, performance anomalies, or unexpected system behavior
- Limitation: May produce false positives and requires human interpretation to confirm actual bugs

Key Difference: Supervised learning is better for detecting known bug patterns, while unsupervised learning excels at finding novel or unexpected issues.

Q3: Why is bias mitigation critical when using AI for user experience personalization?

Answer:

Bias mitigation is critical in AI-driven user experience personalization because:

- **Fairness and Inclusion:** Biased AI can exclude or disadvantage certain user groups based on demographics, leading to discriminatory experiences
- **User Trust:** Users lose confidence in platforms that provide unfair or stereotypical recommendations
- **Legal Compliance:** Biased systems may violate anti-discrimination laws and data protection regulations like GDPR
- **Business Impact:** Bias can alienate customer segments, reducing market reach and revenue
- **Ethical Responsibility:** Companies have a duty to ensure their AI systems treat all users equitably
- **Quality of Service:** Bias reduces the effectiveness of personalization by making incorrect assumptions about user preferences

Without proper bias mitigation, AI personalization can reinforce stereotypes, create filter bubbles, and provide suboptimal experiences for underrepresented groups.

2. Case Study Analysis

Article: AI in DevOps: Automating Deployment Pipelines

Question: How does AIOps improve software deployment efficiency?

Provide two examples.

Answer:

AIOps (AI for IT Operations) significantly improves software deployment efficiency by automating complex processes, predicting issues before they occur, and enabling faster decision-making.

Example 1: Automated Rollback and Deployment Optimization

AIOps tools like Harness use AI to automatically detect failed deployments and roll them back without human intervention. By analyzing historical deployment data, AI predicts potential build failures in advance and optimizes test case execution. For instance, CircleCI uses AI to prioritize test cases based on their historical success and failure rates, running the most critical tests first. This provides developers with faster feedback, reduces deployment time, and minimizes the risk of downtime when new features go live.

Example 2: Proactive Monitoring and Incident Management

AI-powered monitoring tools such as New Relic, Datadog, and Splunk detect anomalies in real-time by analyzing logs, metrics, and traces across systems. Instead of waiting for threshold-based alerts, AI identifies performance degradation patterns before they impact users. For example, Netflix uses AI to monitor streaming components and detect potential issues in real-time, ensuring uninterrupted service. AI also automates incident response by using Natural Language Processing (NLP) to suggest solutions based on previous incidents, resolving issues in seconds rather than hours and significantly reducing downtime.

Overall Impact:

These AI-driven capabilities result in shorter deployment cycles, reduced human error, improved system reliability, and cost efficiency through optimized resource allocation. Organizations can deploy software faster and more confidently while maintaining high availability and performance standards.